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Prediabetes

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Continuing Education Activity

Prediabetes is a precursor before the diagnosis of diabetes mellitus. Adults with prediabetes often may show no signs or symptoms of diabetes but will have blood sugar levels higher than normal. The normal blood glucose level is between 70 mg/dL to 99 mg/dL. In patients with prediabetes, you can expect to see blood glucose levels elevated between 110 mg/dL to - 125 mg/dL. This activity reviews the cause and pathophysiology of prediabetes and stresses the importance of the interprofessional team in its management.

Objectives:

- Identify the etiology of prediabetes.
- Review the evaluation of a patient with prediabetes.
- Outline the treatment and management options available for prediabetes.
- Describe interprofessional team strategies for improving care coordination and outcomes in patients with prediabetes.

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Introduction

Prediabetes is a precursor before the diagnosis of diabetes mellitus. Adults with prediabetes often may show no signs or symptoms of diabetes but will have blood sugar levels higher than normal. The normal blood glucose level is between 70 mg/dL to 99 mg/dL. In patients with prediabetes, you can expect to see blood glucose levels elevated between 110 mg/dL to - 125 mg/dL. However, these levels do not meet the required criteria for a diagnosis of diabetes mellitus. For this reason, many people are not aware that they are living with prediabetes.[1][2][3]

In addition to type 2 diabetes, prediabetes is a risk factor for the development of cardiovascular disease, and stroke. Once diagnosed with prediabetes patients should be checked for progression to type 2 diabetes every one to two years. If screening is negative for prediabetes, repeat screening should be carried out every 3 years as per the United States Preventive Services Task Force (USPSTF). Lifestyle changes through improved nutrition and physical activity are the firstline treatment for preventing the transition from prediabetes to diabetes which can be as high as 70%.

Etiology

The following factors put the patient at greater risk:

- Overweight or obesity (especially a body mass index (BMI) greater than 25 kg/m^2)
- Family history of diabetes mellitus (parent or sibling)
- Diabetes during pregnancy (gestational diabetes)
- High-risk ethnic groups: African American, Latin America, Native American, or Asian/Pacific Islander

- Hypertension
- Physical inactivity
- Dyslipidemia with levels of HDL cholesterol less than 40 mg/dL (men) or less than 50 mg/dL (women) or triglycerides more than 250 mg/dL
- Polycystic Ovarian Syndrome

Epidemiology

According to the Center for Disease Control and Prevention, about 84 million American adults are currently facing prediabetes. This equals one in three adults in America. About 90% of these adults do not know that they are currently living with prediabetes and setting themselves up for all the implications this entails. The incidence of diabetes is evidently growing at rapid rates globally. In America alone, about 1.5 million Americans are being diagnosed with diabetes every year. These increases are parallel with the rapid increases in the prevalence of obesity. Annually, diabetes remains the seventh cause of death in the United States and is currently costs about $245 billion in the United States. Due to this, preventing this trending progression should be at the top of the list as a national health focus and strategy. The focus on management and diagnostic studies should come second given that this disease is preventable.[4][5][6]

Pathophysiology

Since prediabetes is the precursor for diabetes mellitus, the pathophysiology is relatable. Hyperglycemia will cause production and release of insulin by the pancreatic beta cells. Excess insulin exposure for long periods of time diminishes the reponse of the insulin receptors the function of which is to open glucose channels leading to entry of glucose into the cells. Decreased function of the insulin receptors leads to further hyperglycemia further perpetuating the metabolic disturbance and leading to the development of not only diabetes type 2 but also metabolic syndrome. In prediabetes, this process is not to the extent of diabetes mellitus but is a first step in a metabolic cascade which has potentially dangerous consequences if not adequately addressed. Hence its imperative to start treatment at the earliest.[7] If treatment is not started or if the treatment is not adequate, adverse effects on large and small blood vessels (e.g. arteries of the cardiovascular system or retina, kidney, and nerves) may occure.

History and Physical

In majority of the patients with prediabets do not experience any symptoms and hence appropriate screening and monitoring especially in individuals with family history is needed. In the minority of patients who do experience symptoms, they can be as follows:

1. Increased appetite
2. Unexplained weight loss/weight gain
3. High BMI
4. Weakness
5. Fatigue
6. Sweating
7. Blurred vision
8. Slow healing cuts or bruises
9. Recurrent skin infections/gum bleeding

The single sign of prediabetes is elevated blood glucose on a blood test that is not high enough to be classified as type 2 diabetes mellitus.

Evaluation

The following tests can be used to screen for prediabetes:

- 12 hour Fasting blood glucose levels: Blood glucose levels fall between 100 mg/dL to 125 mg/dL, it is diagnostic of prediabetes.
- Two-hour glucose tolerance test: this test will measure blood glucose levels before and after ingestion of 75 g of glucose solution; if the test shows blood glucose levels that fall between 140 mg/dL to 199 mg/dL, it is diagnostic of prediabetes.
- A glycated hemoglobin test (also known as hemoglobin A1C) measures the average blood glucose level over the last 2 to 3 months. If it falls between 5.7% and 6.4%, it is diagnostic of prediabetes.
- A random plasma glucose test measures blood glucose levels at any time; if the blood glucose levels fall between 140 mg/dL to -199 mg/dL, it may be indicative of prediabetes. This test will require a follow-up test to be accurate.

Screening should start between ages 30 to 45 and repeated at least every 3 years. In high-risk patients, you can initiate screening earlier and follow-up more frequently.[8][9]

Treatment / Management

The most important management in prediabetes is a lifestyle change and promotion of intense weight loss. Reducing weight by 7% through a low-fat diet, in addition to an exercise regimen of about 30 minutes per day, is the overall goal of management. [10][11][12]

Approximately 70% of people with prediabetes will go on to be diagnosed with diabetes mellitus. However, this is not inevitable. Prediabetes managed appropriately can prevent diabetes mellitus and lower the risk of cardiovascular disease.

Some patients will need to take some medications. These patients include those that have failed to maintain adequate lifestyle therapy or are at high-risk for developing type 2 diabetes. The most common medications used for prediabetes are metformin and acarbose, which will help prevent the development of diabetes mellitus. These two drugs have minimal side effects and work well in prediabetic patients.

Differential Diagnosis

- Type 1 or type 2 diabetes
- Insulin resistance
- Metabolic syndrome

Complications

- End stage renal disease
- Blindness
- Hypertension
- Insulin resistance
- Peripheral neuropathy
- Limb loss

Pearls and Other Issues

Prevention is the key of prediabetes. The best preventative measures are:

- Maintaining a diet rich in fiber

- Exercising regularly
- Losing weight
- Adhering to medications prescribed by your doctor
- Smoking cessation

Many studies suggest that a low-carbohydrate diet can help control insulin resistance, blood glucose levels, and weight issues. Consuming low sodium at levels less than 1500 mg per day, limiting alcohol to zero or one drink per day, and cutting out added sugar and unhealthy fats will also help prevent prediabetes from developing. Prediabetes is reversible and can only be managed by making these significant lifestyle changes and having physicians who know how to educate patients on adopting healthier lifestyle habits.

Enhancing Healthcare Team Outcomes

Today a great deal of effort is based on reversing the prediabetic state. This is best done in an interprofessional fashion that involves an endocrinologist, bariatric surgeon, dietitian, pharmacists, weight loss nurse, and a physical therapist. The patient should be educated on the importance of exercise and discontinuation of smoking. Further, the blood pressure must be well controlled and the hyperlipidemia lowered. The patient must be educated on the importance of eating a healthy diet and remaining compliant with medications to lower the blood glucose and cholesterol. [13][14](Level V)

Outcomes

Many studies have sown that there is a relationship between persistently elevated blood glucose levels and risk for adverse cardiac events and death. Evidence shows that individuals with prediabetes are susceptible to many metabolic complications that may lead to blindness, stroke, heart disease, and renal failure. These individuals are also at a high risk for developing peripheral neuropathy and loss of limb. Further, the elevation of blood glucose during pregnancy also increase the risk of maternal and fetal mortality. [15][4](Level V)

Review Questions

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